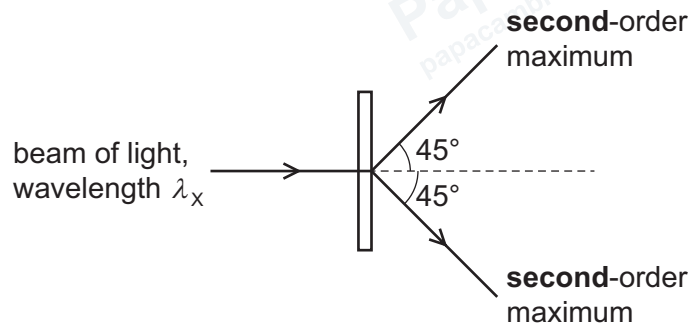


- 27 The diagram shows a beam of light of wavelength λ_x that is incident normally on a diffraction grating.



The diagram shows only the **second-order** intensity maxima that are produced at an angle of 45° to the direction of the incident beam of light.

A replacement beam of light of a different wavelength λ_y is now incident normally on the diffraction grating. This new wavelength also produces intensity maxima that are at an angle of 45° to the direction of the incident beam of light.

What is a possible value of the ratio $\frac{\lambda_x}{\lambda_y}$?

- A** 0.25 **B** 0.75 **C** 1.25 **D** 1.50